Final Project

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The objective of the final project is to control the robot via LLM/VLM and an "agentic AI framework".

The user should be able to type/speak the prompt, and the robot should execute it.

Sample prompts (other variations are possible):

1. Pick up the small blue block and place it in the box on the right.

2. Pick up the large yellow block and place it in the box on the left.

3. Pick up a small block and place it on top of the large block.

4. Pick up the small blue block, rotate by 90 degrees in z, and place it on large red block.

5. Pick a small yellow block and place it to the right of the red block.

Note: You need to ensure that the "AI agent" is kinematically aware, i.e., the robot could be at any height relative to the blocks. Use "monocular depth estimation algorithms" to estimate the relative depth between the object and the robot.

You should train the robot for as many prompts as possible.

Thanks,

Sangram

1. Demo (50 points) is required along with the report (25 points), code (10 points), and video (15 points).

Points	100
Submitting	a file upload

Due	For	Available from	Until
Dec 13, 2025	Everyone	-	Dec 14, 2025 at 11:59pm

+ Rubric